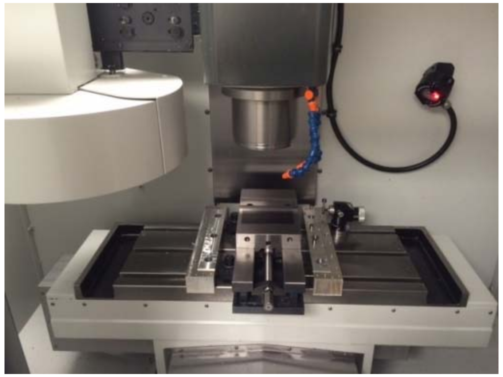
Advanced Manufacturing Technology & Prototyping (MDP494)

Lecture 1_Part 2

Introduction to CNC Milling programming (part 1)



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Milling Machine

CNC MACHINES VERSUS MANUAL MACHINES

Differences

- **Accuracy:** In conventional machines the operator have to adjust the dimensions for each workpiece therefore it is not accurate every time. While in CNC machines the dimensions are set once and can be repeated with the same accuracy every time.

- **Time:** CNC machines are faster in machining since it changes the tool automatically (can use up to 24 tools) while in conventional machines the whole machine is stopped to change tool manually (one tool at the time).

Differences, cont.

- **Quality.**

- **Gear box**: In conventional machines there is a gear box which determines the speed therefore it have a limited number of speed while in CNC machines it is an electric motor therefore it can reach infinite number of speeds within a certain range.

- **Repeatability:** fixing errors in CNC machines that won't be repeated while in conventional the measurements are done using a Vernier.

GENERAL INFORMATION ABOUT THE MILLING MACHINE

Information

- The CNC machine is a 3 axes (moves in x, y and z) machine with control program i.e. Sinumerik with Alpha numeric as language of the code.

- Each CNC machine have three main parts: The machine, the control unit and the program of instructions.

- There are other machines that have higher number of DOF e.g. 12 DOF which are divided into 6 DOF by the tool and 6 DOF by the table.

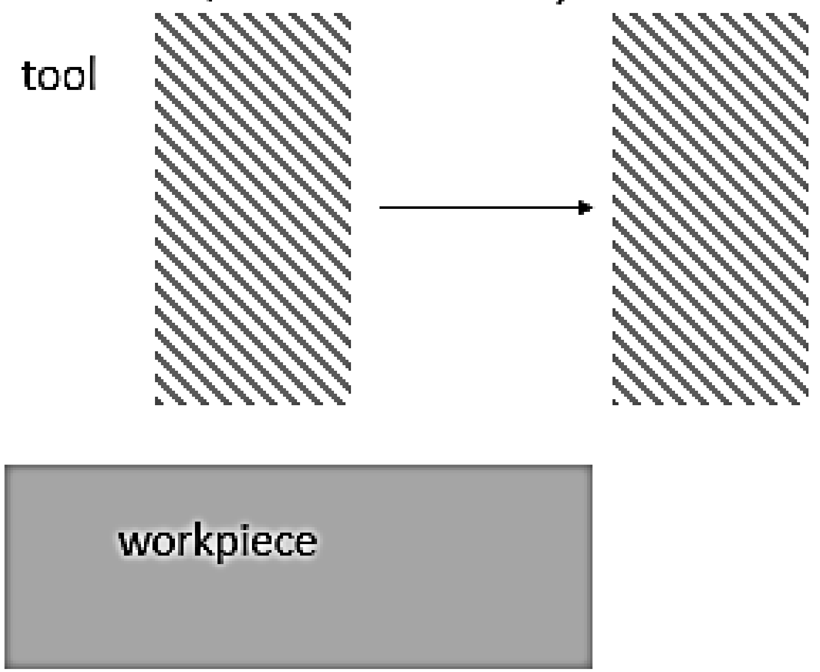
Information, cont.

- Coolant or air used according to the type of material.
- Vice is a clamp by pressure, not moving.
- Position given in x and y at center of the tool while in z at the tip of the tool measured from the machine zero.
- Zero position when x, y and z are equal to zero.

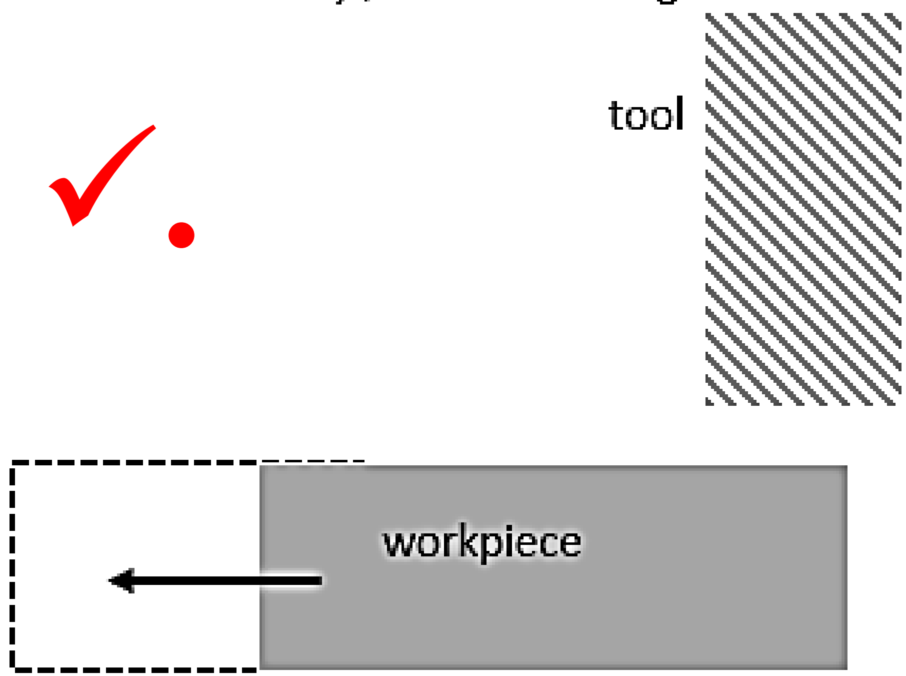
Information, cont.

- The motion should be relative to the motion of the tool. If tool is to be moved in positive x-axis direction; there are two cases:

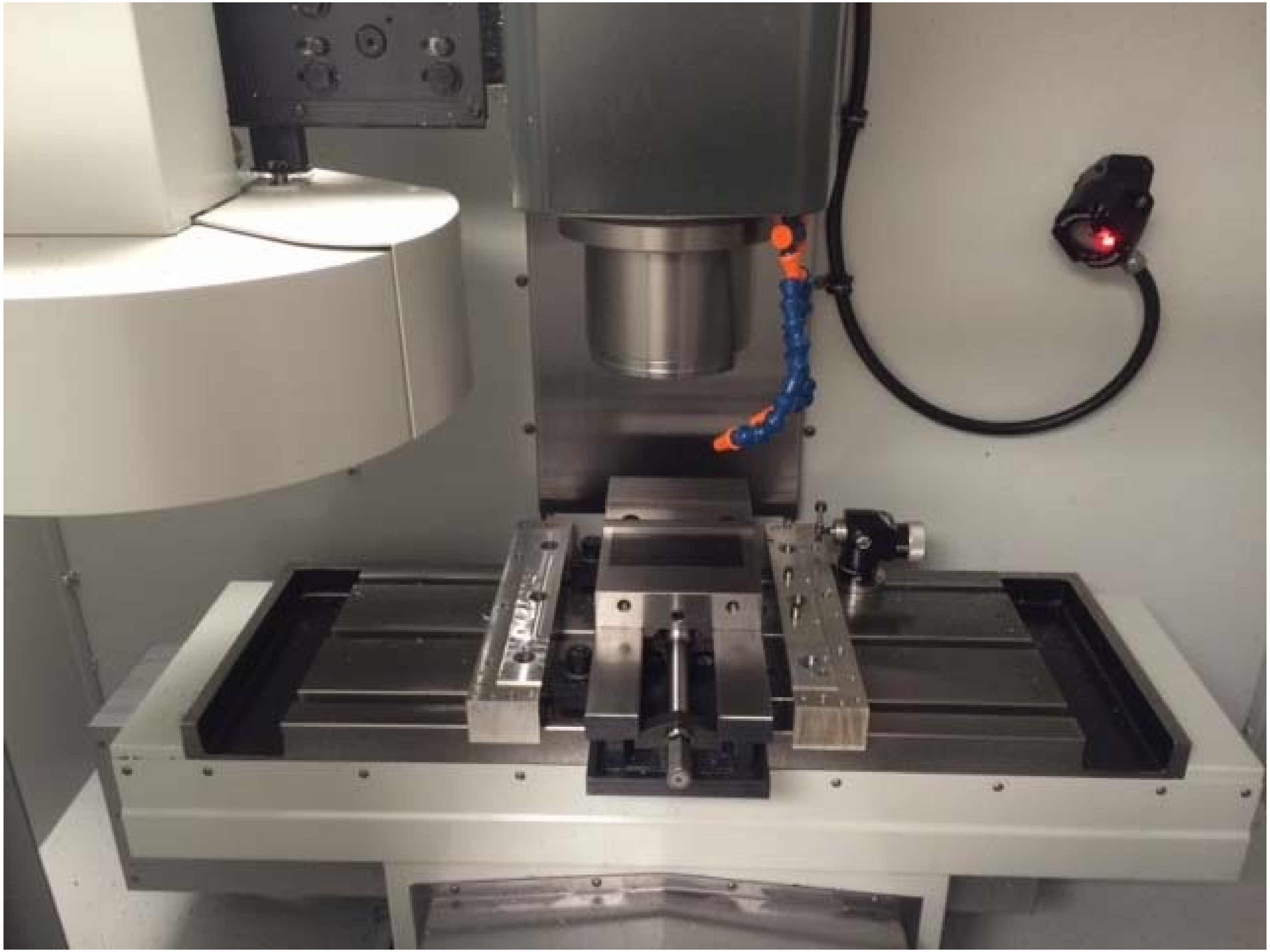
If tool is moving in positive direction, tool can move, table is stationary.



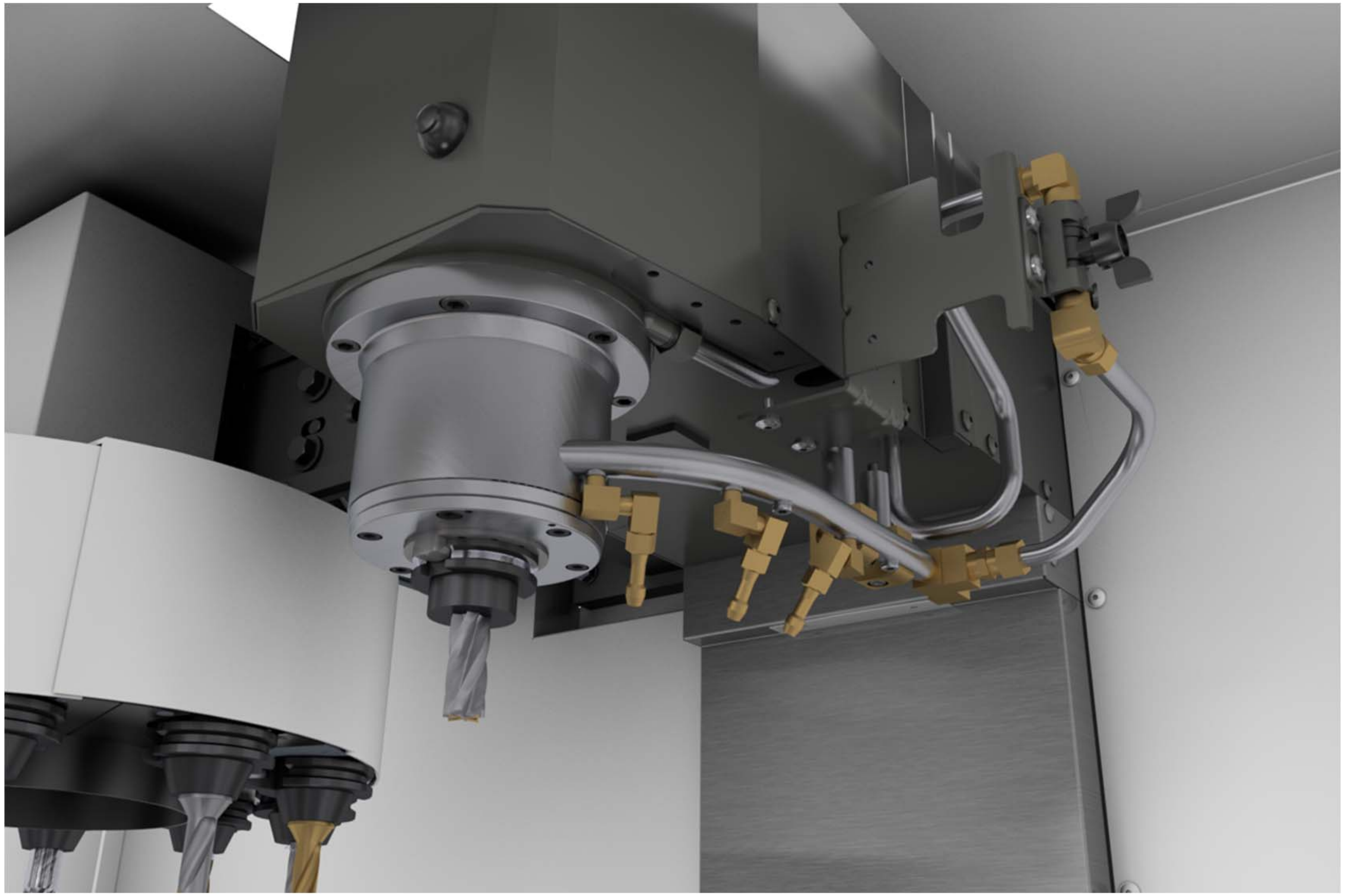
If tool is to move in positive direction , tool is stationary , table is moving to left.



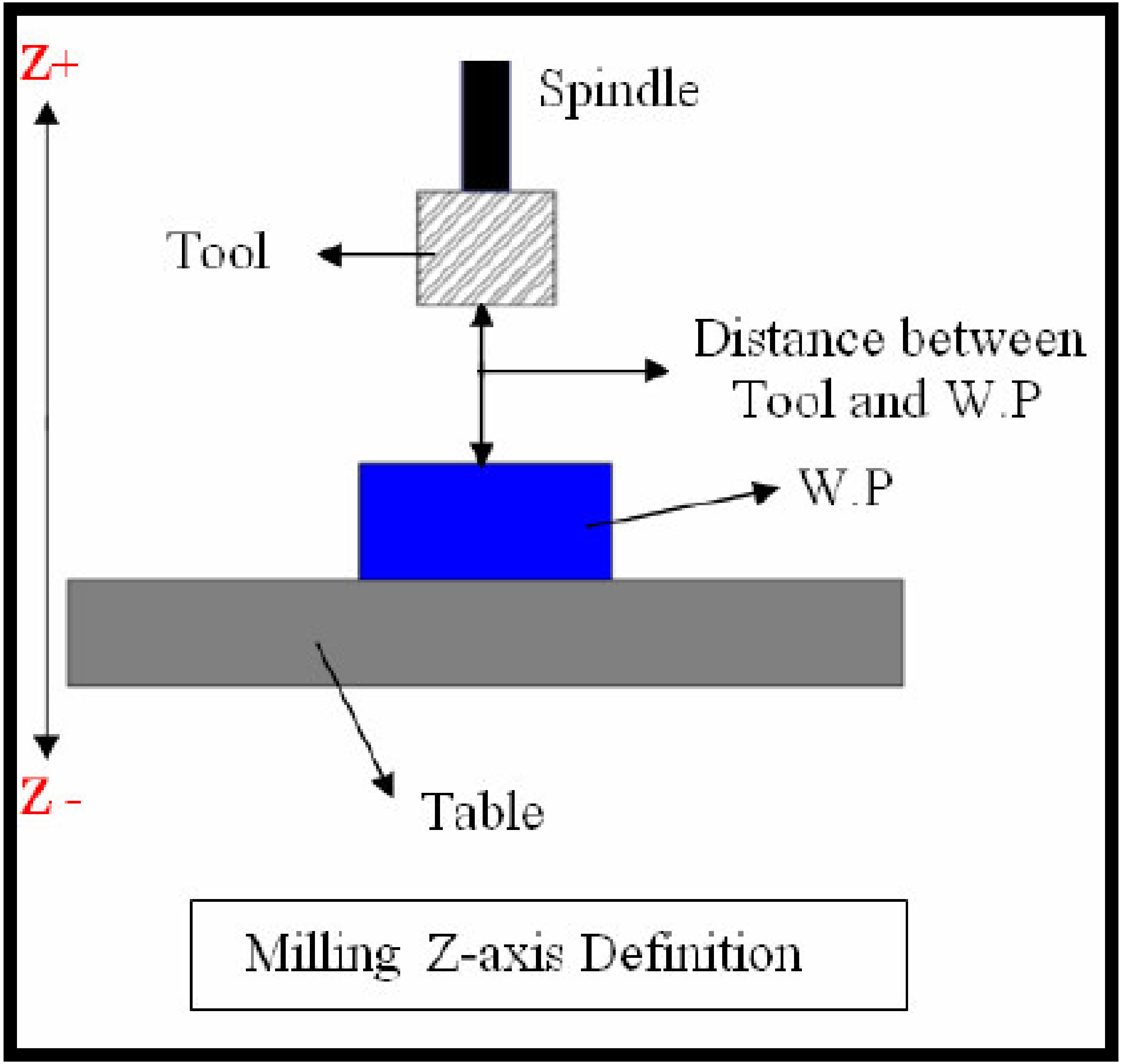
CNC Milling Machine



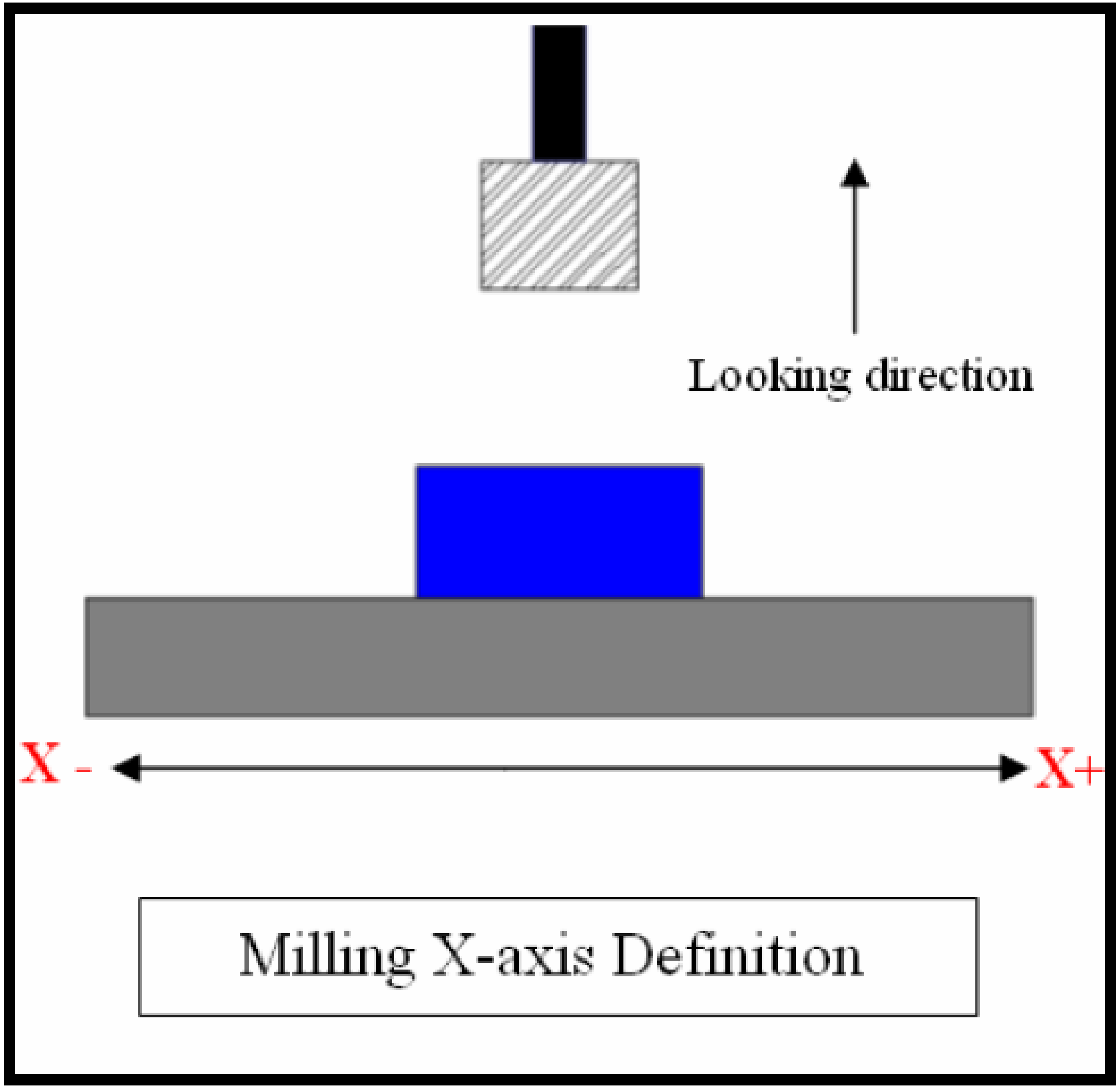
Spindle

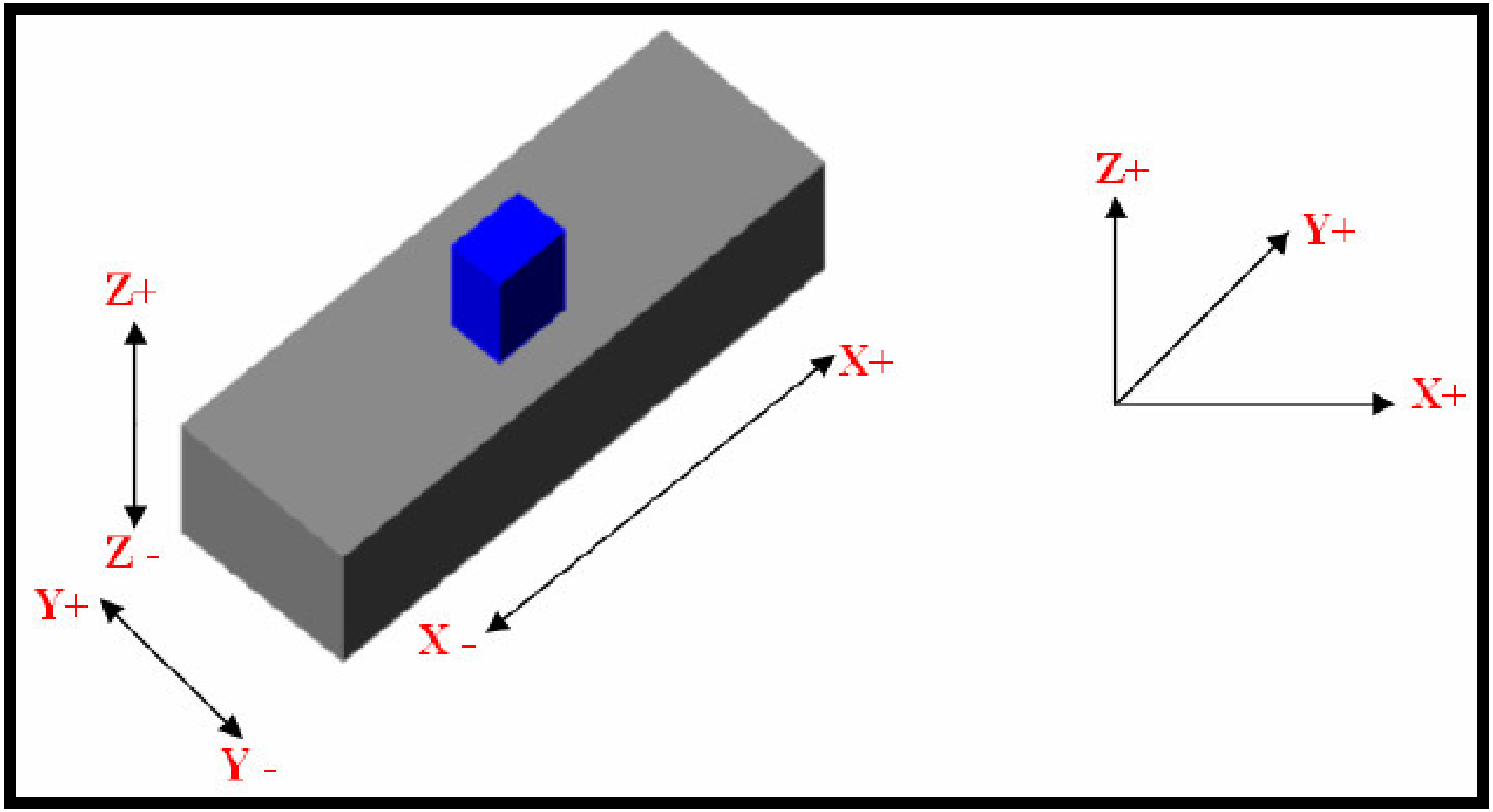


CNC AXIS DEFINITION

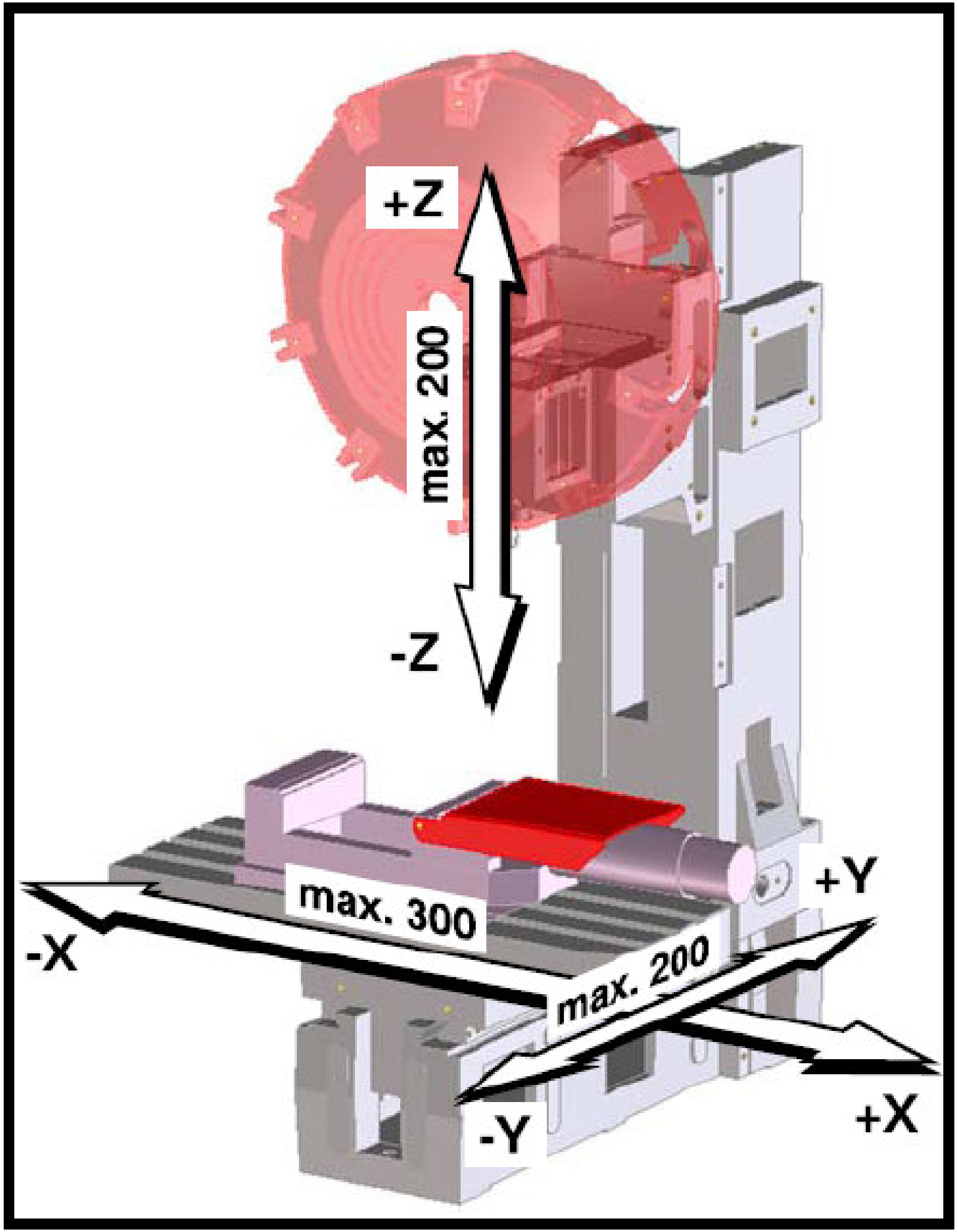


Milling Z-axis Definition

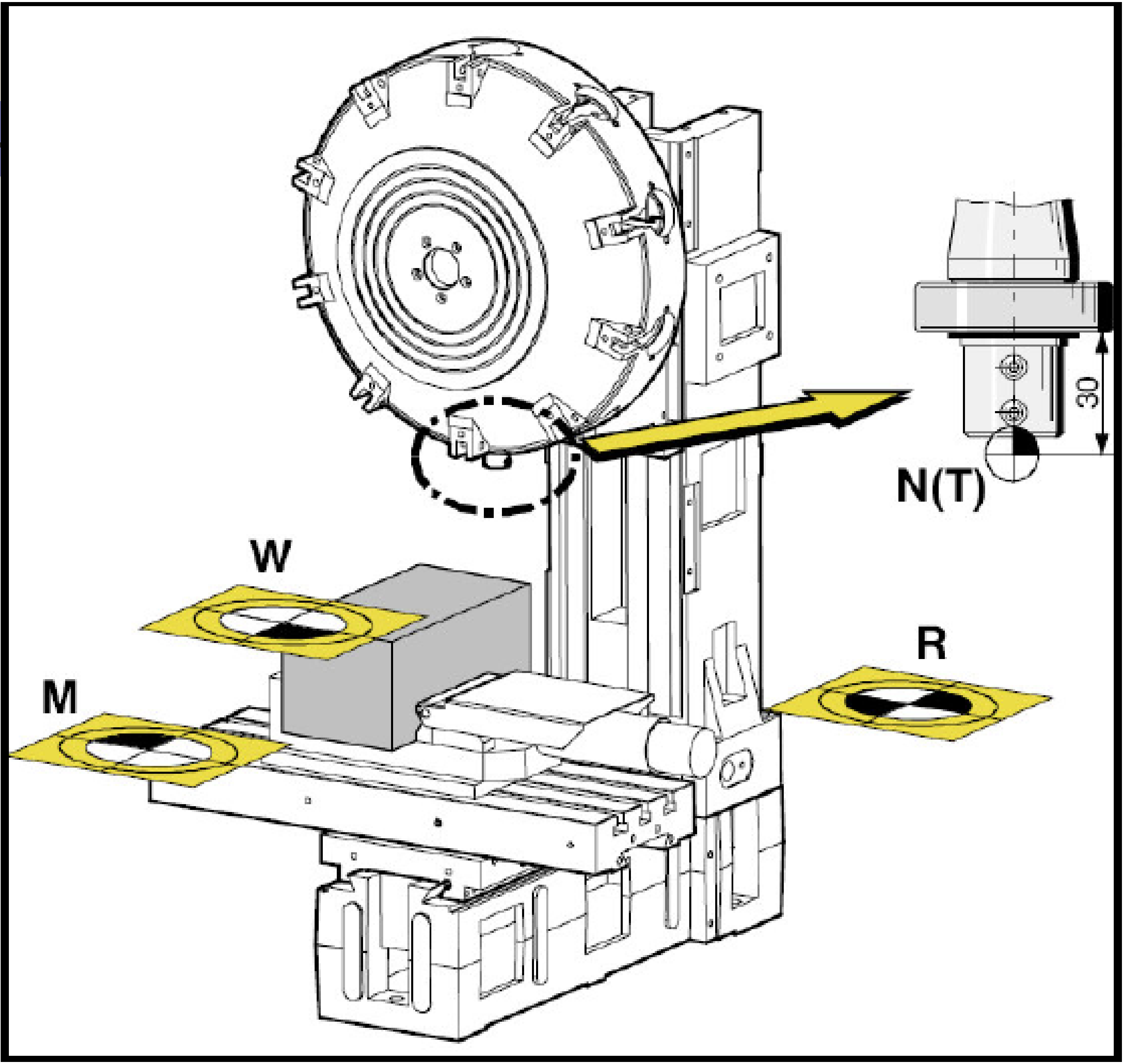

Looking direction
X -
X+
Milling X-axis Definition



Z+
Z -
Y+
Y -
X+
X –
Z+
Y+
X+


+Z
max. 200
-Z
max. 300
-X
+Y
max. 200
-Y
+X

MACHINE POINTS

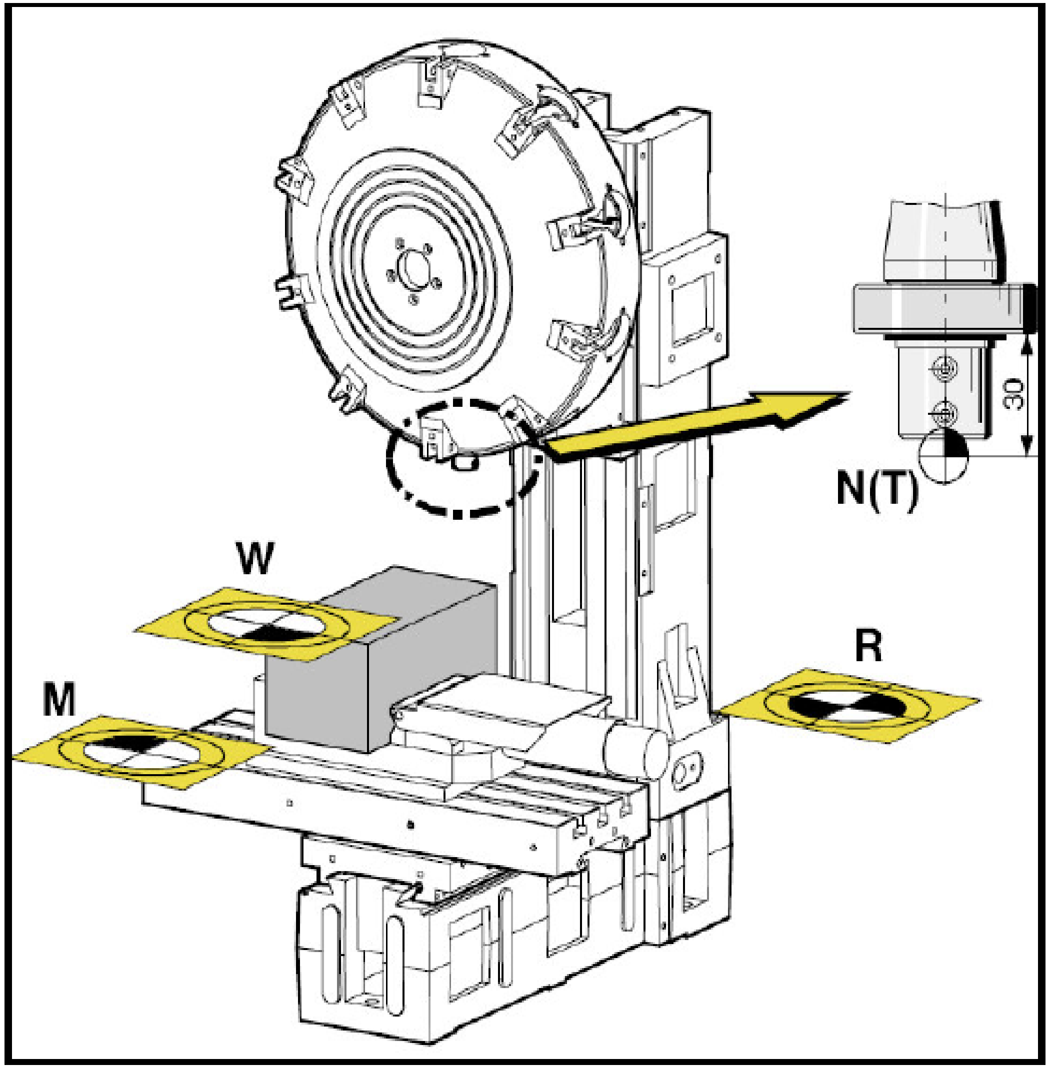


W
M
R
N(T)
30

Machine Points

Reference point R

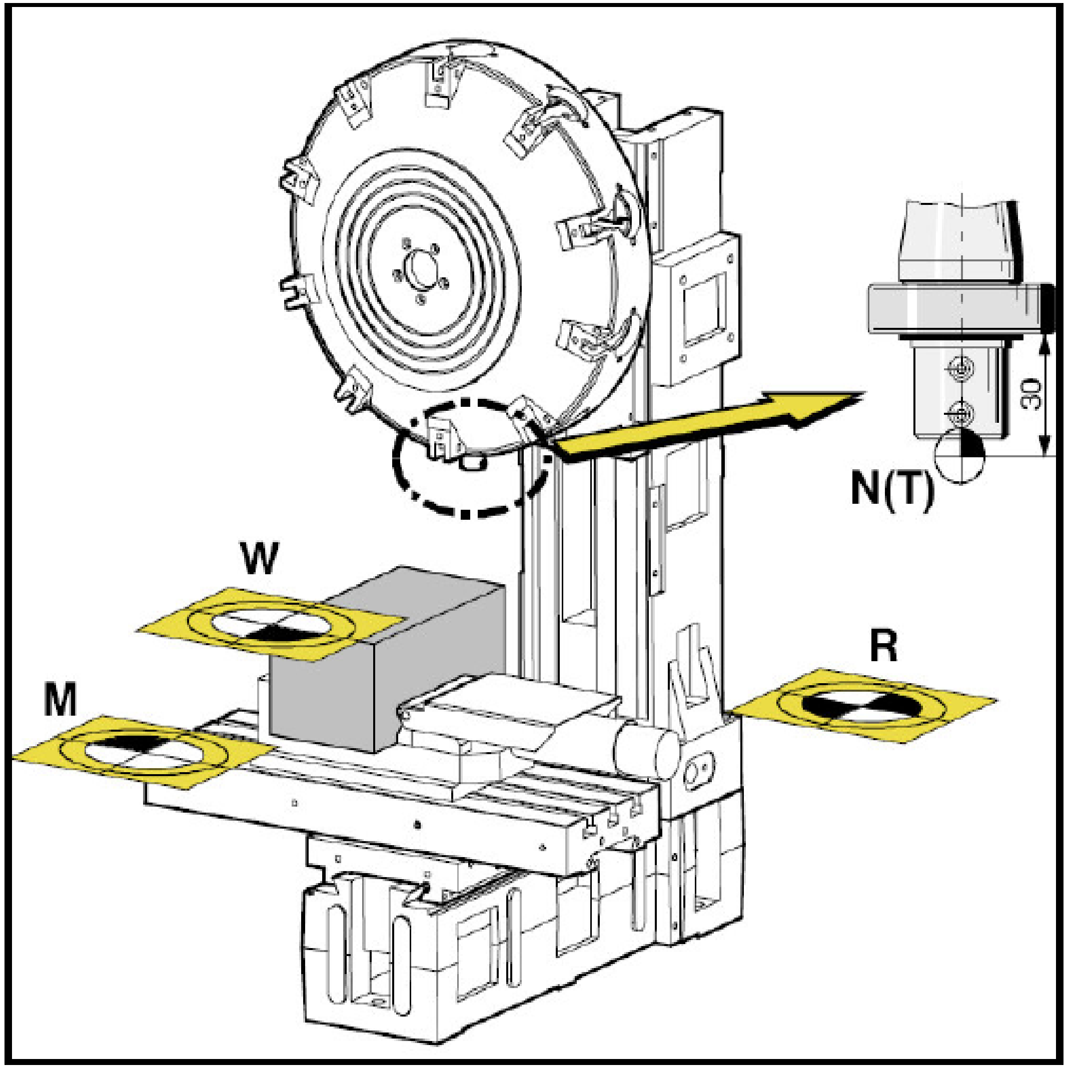
- The reference point R is a fixed point at the machine.
- It serves for the calibration of the measuring system.



Machine Points, cont.

Machine Zero Point M

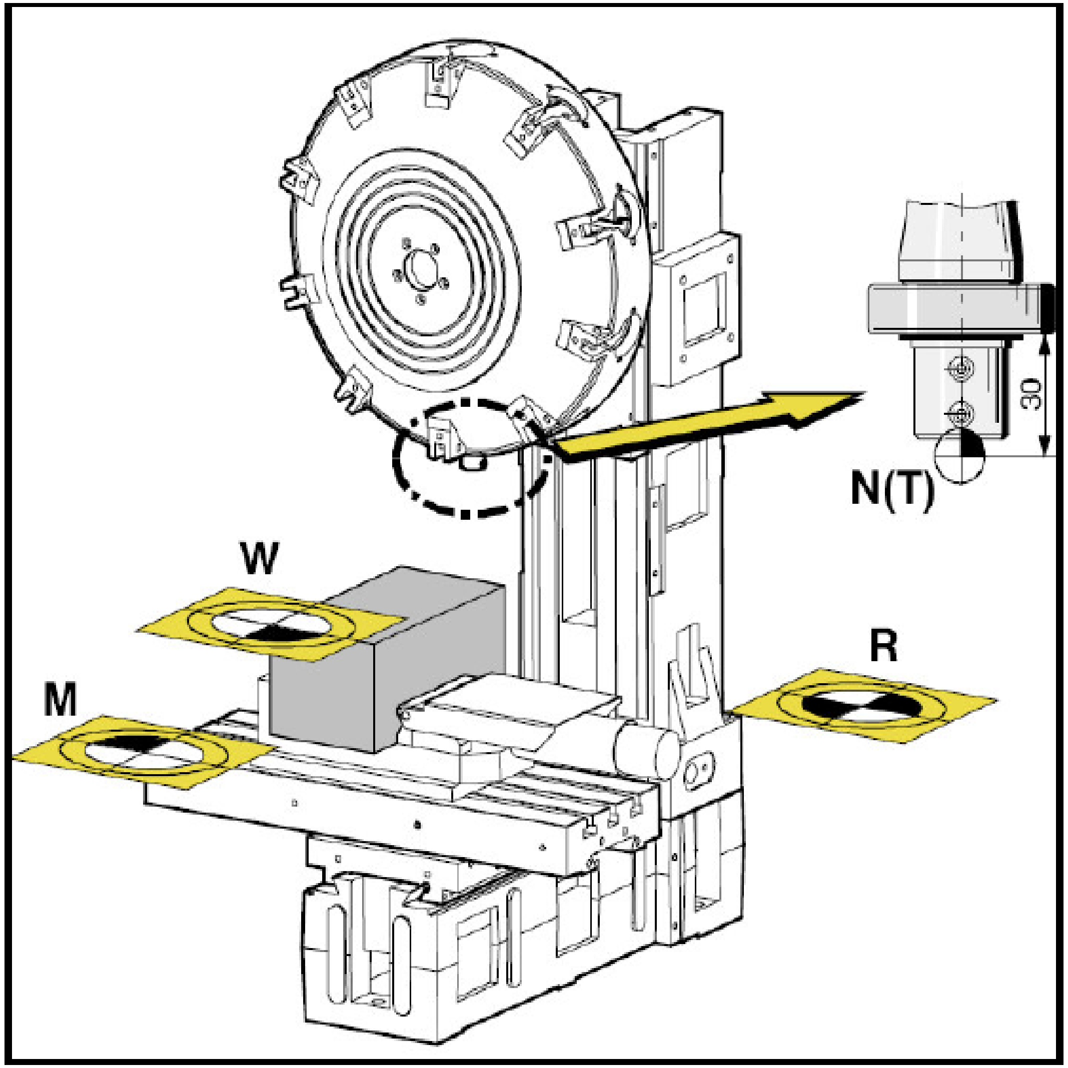
- The machine zero point M lies on the surface of the milling table on the left front corner.
- The machine zero point M is the origin of the coordinate system.



Machine Points, cont.

Workpiece Zero Point W

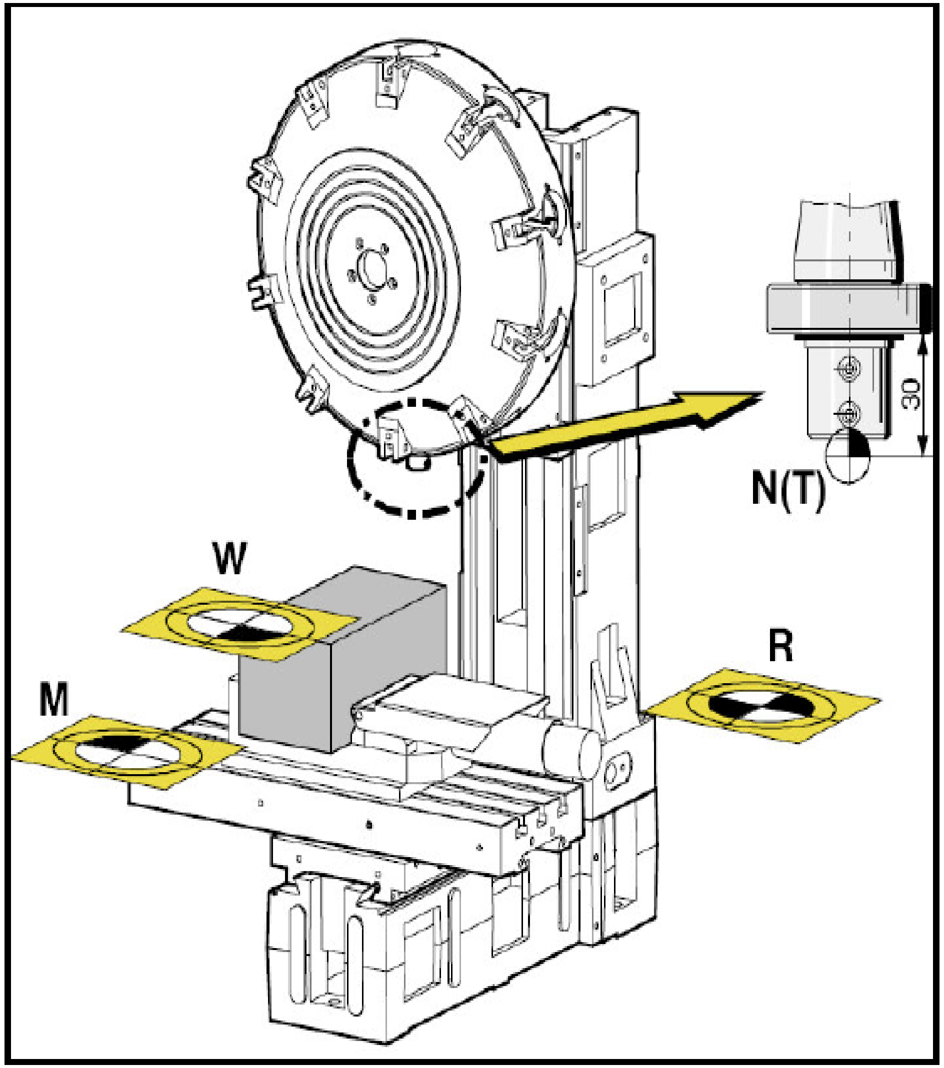
- The workpiece zero point W can be freely programmed by the user.
- By programming a wp zero point W the origin of the coordinate system is displaced from the machine zero point M into wp zero point W



Machine Points, cont.

Tool holding-fixture reference point N(T)

- **The tool holding fixture reference point N(T) lies exactly in the rotary axis of the milling spindle, in a distance of 30 mm in Z-direction from the shoulder of the ball bearing of the tool holder.**
- **The tool lengths are described starting from this point.**



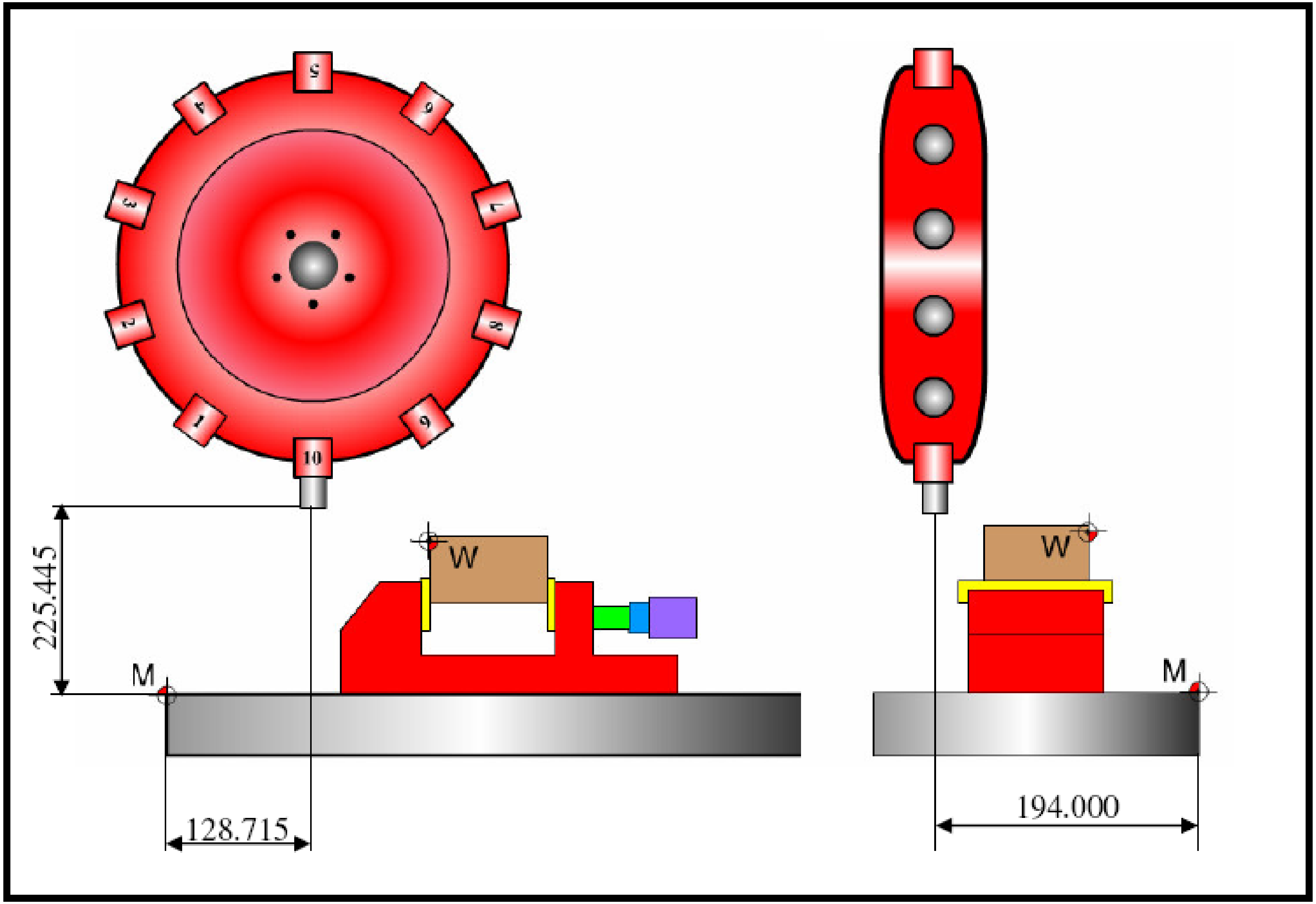
ZERO OFFSET

Zero offset

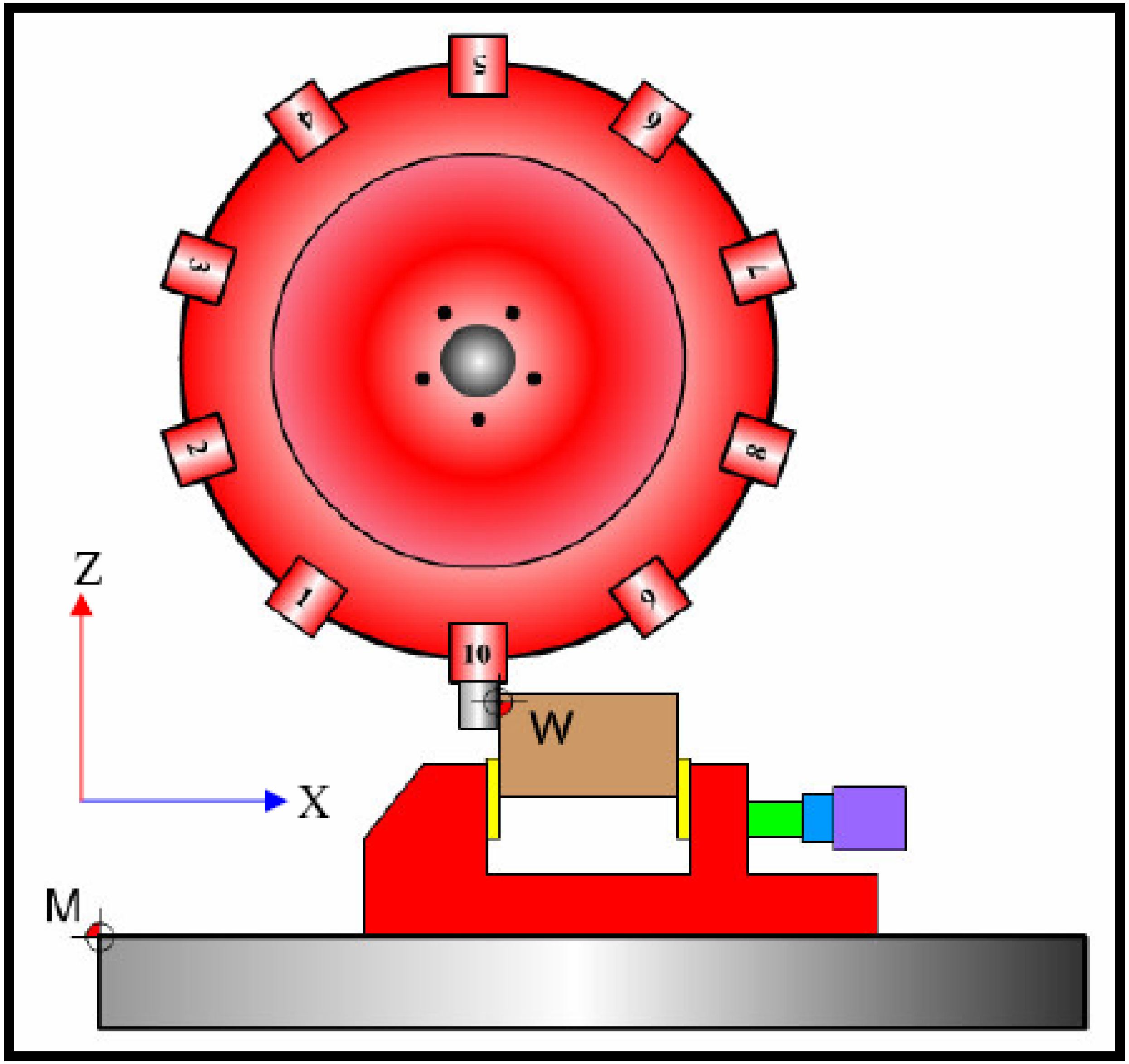
- The shown coordinates are the distances between the machine zero and the tool zero in X,Y and Z

WCS	Position		Dist-to-go
X	128.715	mm	0.000
Y	194.000	mm	0.000
Z	225.445	mm	0.000
C	12.958	deg	0.000

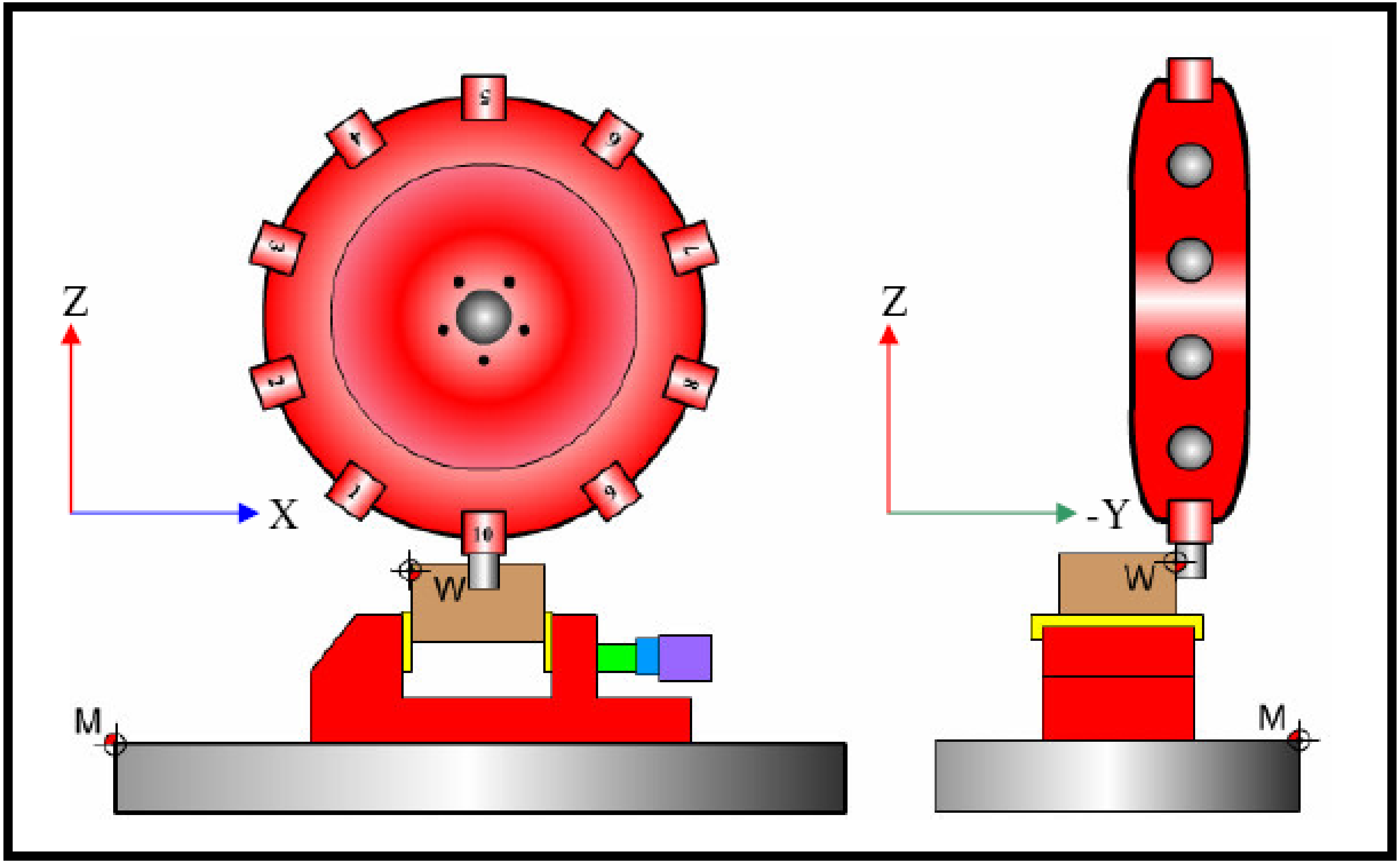
Coordinates



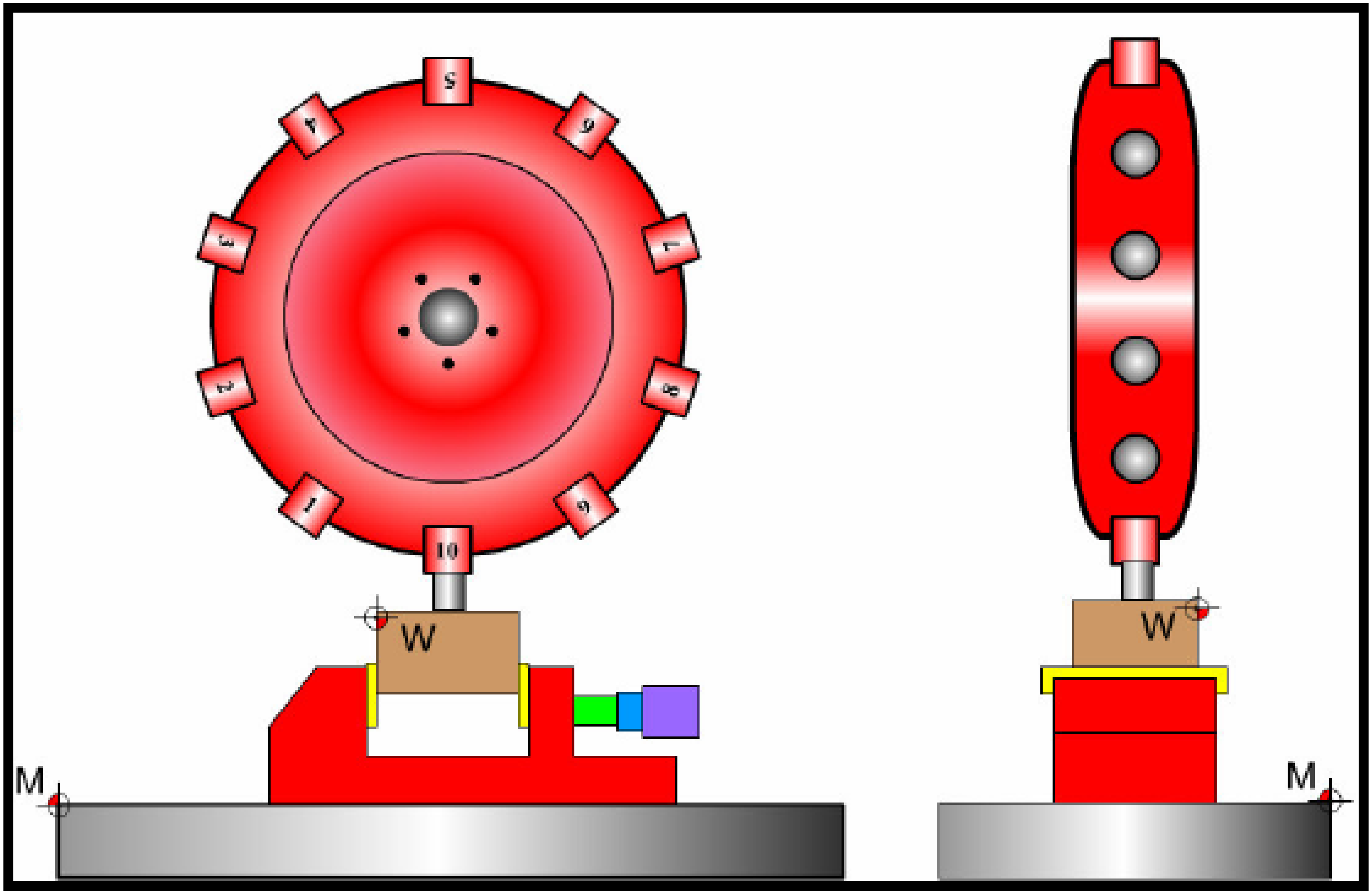
Zero offset in x



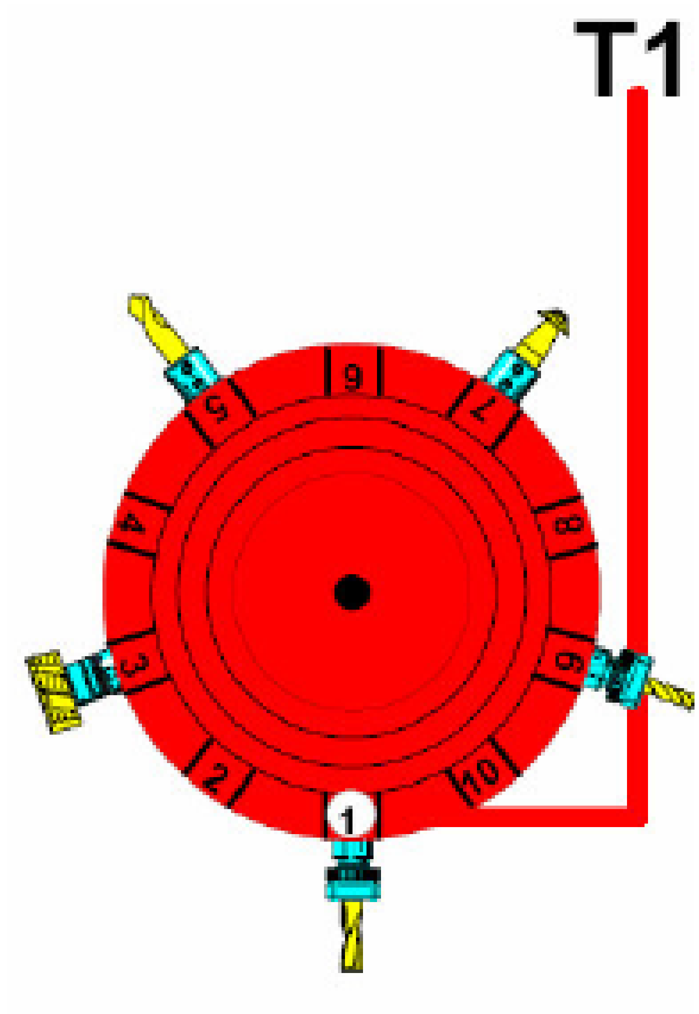
Zero offset in Y

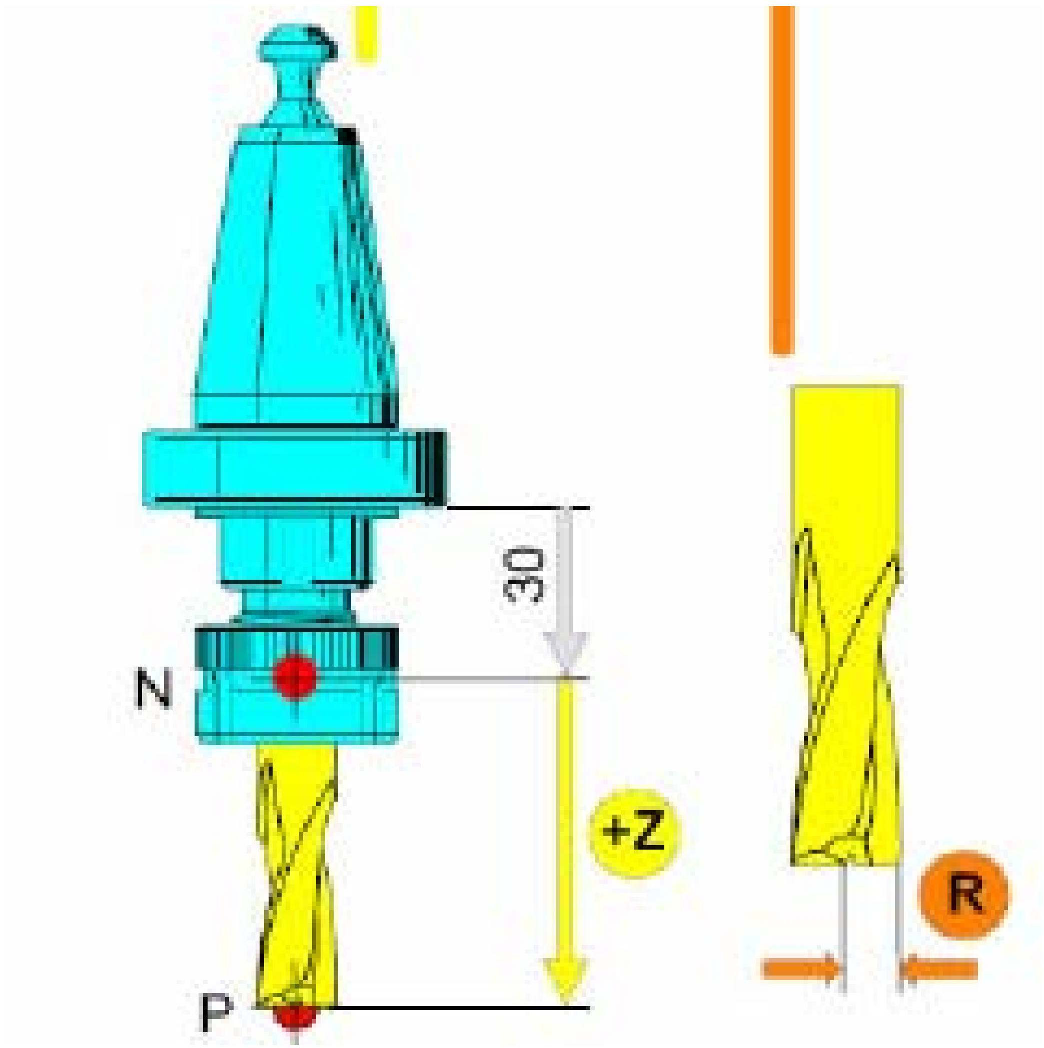


Zero offset in Z

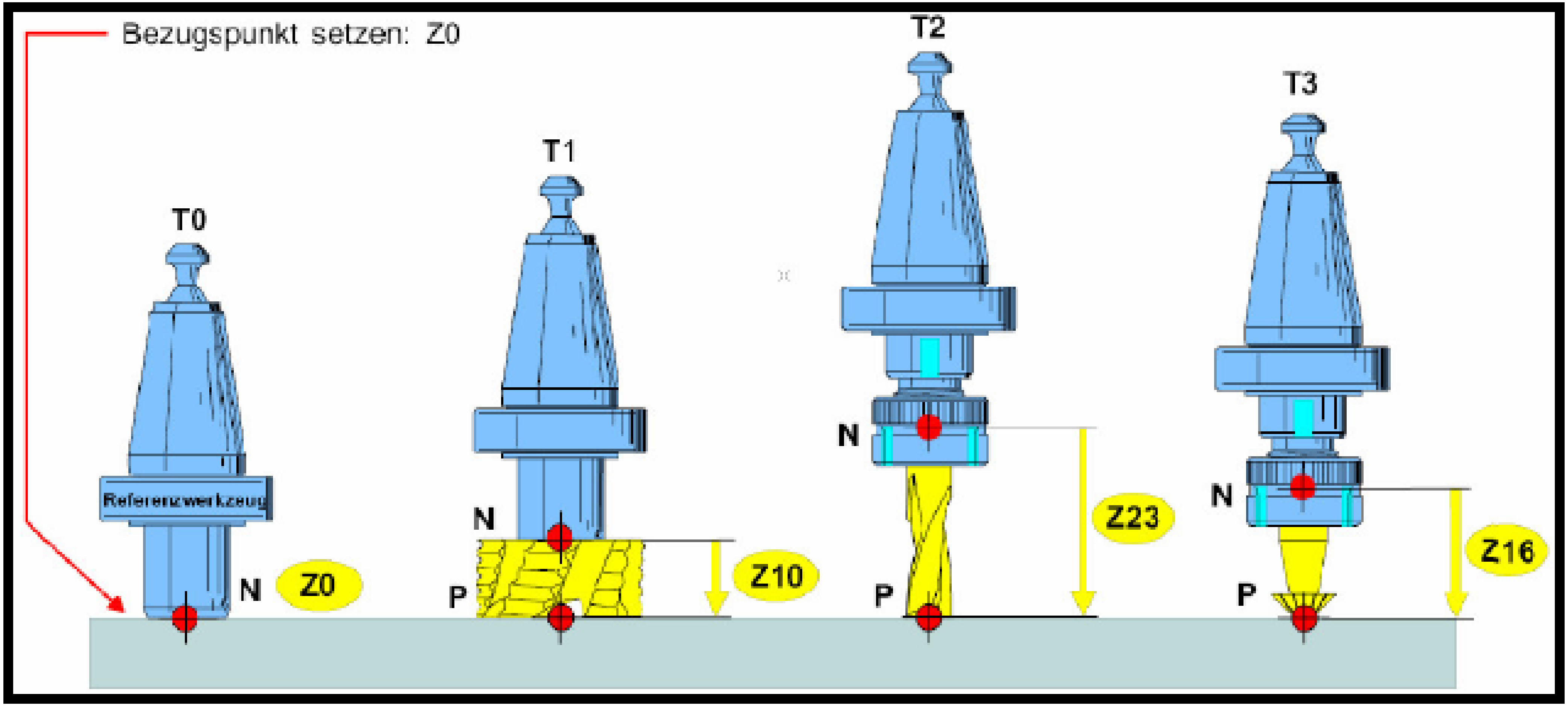


Tool offset





Tool offset



Bezugspunkt setzen: Z0
T0
Referenzwerkzeug
N
Z0
T1
N
P
Z10
T2
N
P
Z23
T3
N
P
Z16

TERMINOLOGIES

Axes digits

- **X, Y, Z ± 3.2:**

3 means that we can use 3 digits to define the lengths, e.g. 100, 450, 312…..so on

2 means that we can use 2 digits to the accuracy (0.01), e.g. 100.43, 450.31, 312. 99 ……… so on

S: Speed of Spindle

- **S2, S3, S4:**

S2: means we can use 2 digits to define the speed of the spindle.

Ex. S70

Means the speed of the spindle is 70 rev/min

S3: means we can use 3 digits to define the speed of the spindle.

Ex. S1000, S2000

F: Feed

F000: **is in mm/min** **inch/min**

mm/rev **inch/rev**

Ex. F100

Means the feed of the spindle is 100 mm/min

T: Tool

- **T means the tool number**
- **T0: T1, T2, T3,……….,T16, …….**

N: Sequence number

- **N is the sequence number in the main program**
- **N05, N10, N15, N20,………., N140, ……..**

G: Motion function

G is the operation name which is written in the operation panel of the machine.

- Its followed by two digits from 0 to 99 so that there is 100 of G in the machine program.

M: Miscellaneous Function

- It is an auxiliary functions used to define un-cutting operations.
- There are 100 of M function M00-M99

Programming steps

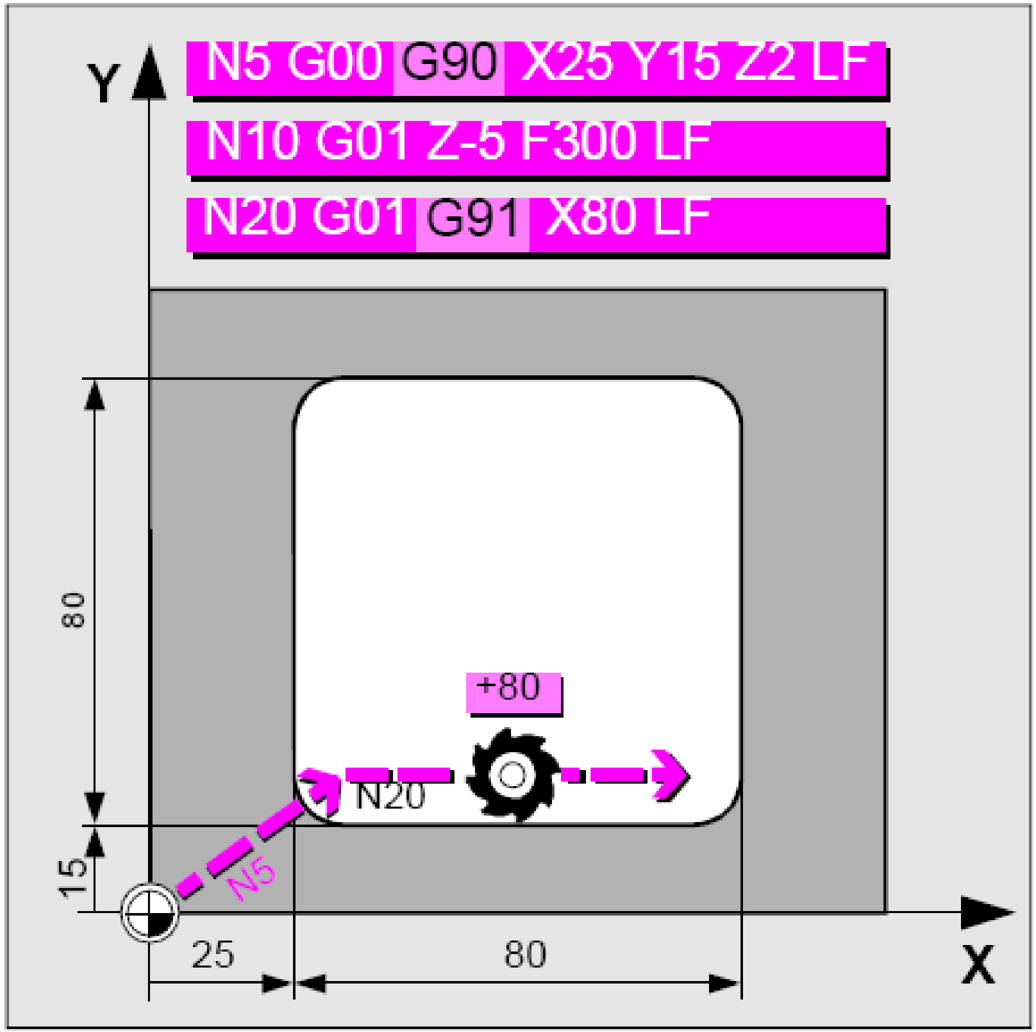
1- BEGINNING OF THE PROGRAM

1- Beginning of the program

- **Absolute and incremental dimensions, G90, G91**
- **Zero offset, G54 to G57**
- **Selection of working plane G17 to G19**
- **Workpiece dimensioning**

Beginning of the program, cont.

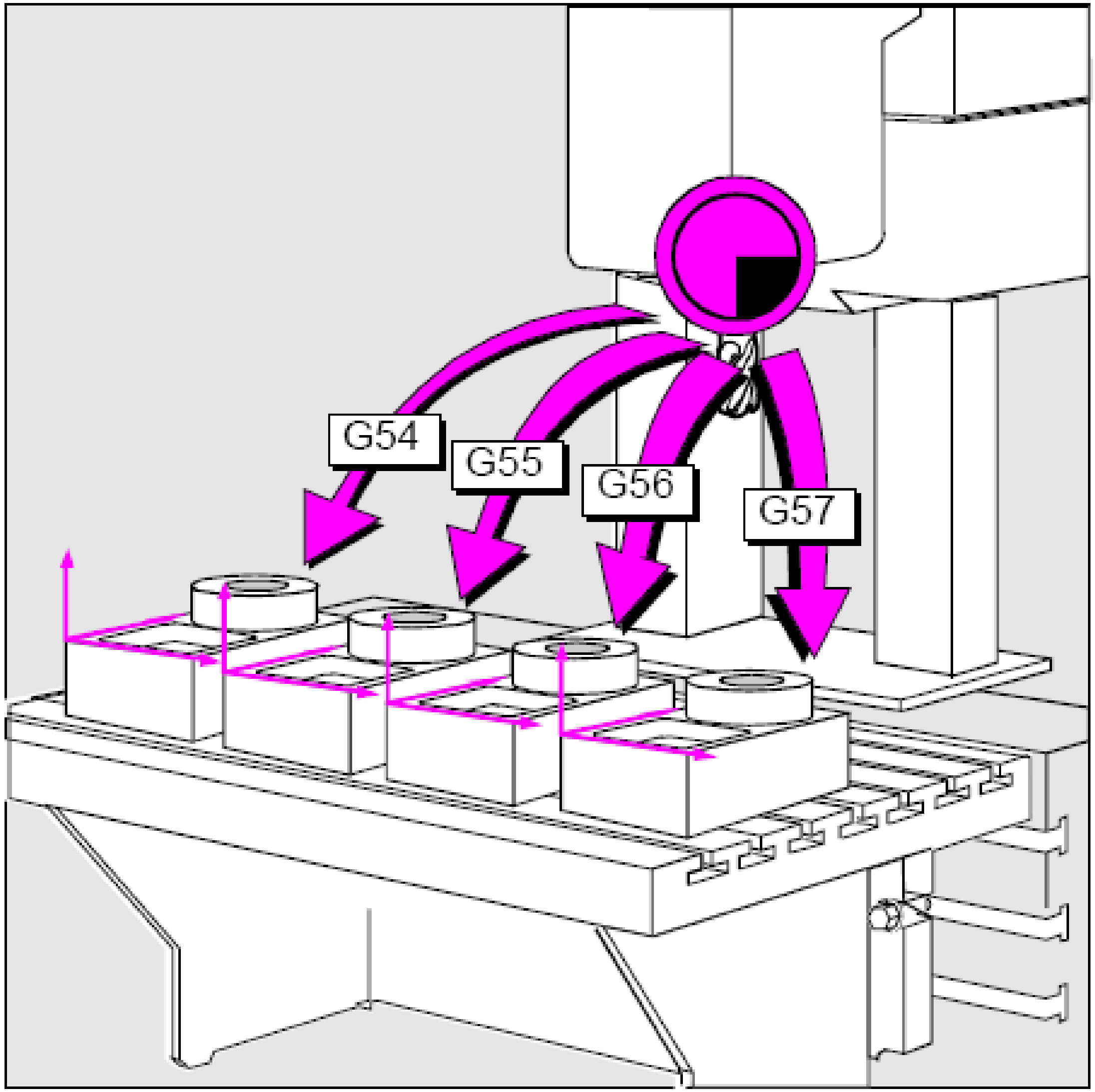
- **G90: Absolute dimension input, all data refers to the actual workpiece zero.**
- **G91: Incremental dimension input, each dimension refers to the contour point last input.**



Beginning of the program, cont.

Zero offset:

- **G54 to G57 :**

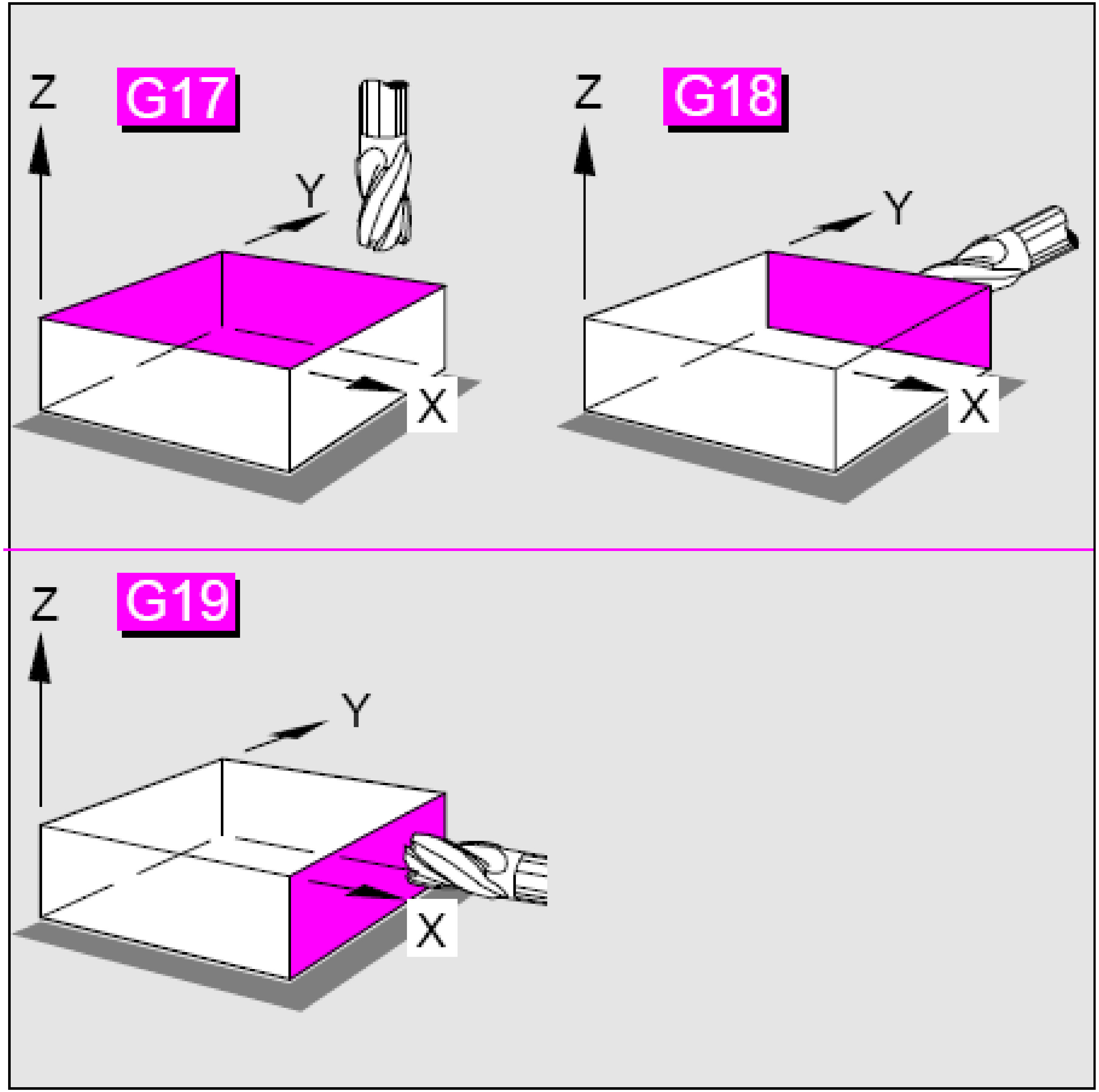


Zero offsets make multiple machining operations possible

Beginning of the program, cont.

- **Selection of working plane G17 to G19.**

Command	Working plane	Infeed axis
G17	X/Y	Z
G18	Z/X	Y
G19	Y/Z	X



Selection of working planes for horizontal and vertical milling operations

Workpiece dimensioning

- **G70: Input system inch**

- **G71: Input system metric**

END